Sentiment Analysis-Based Study on Chinese Audiences' Acceptance of "Friends"

STAT 2630SEF Group Project

Group 8

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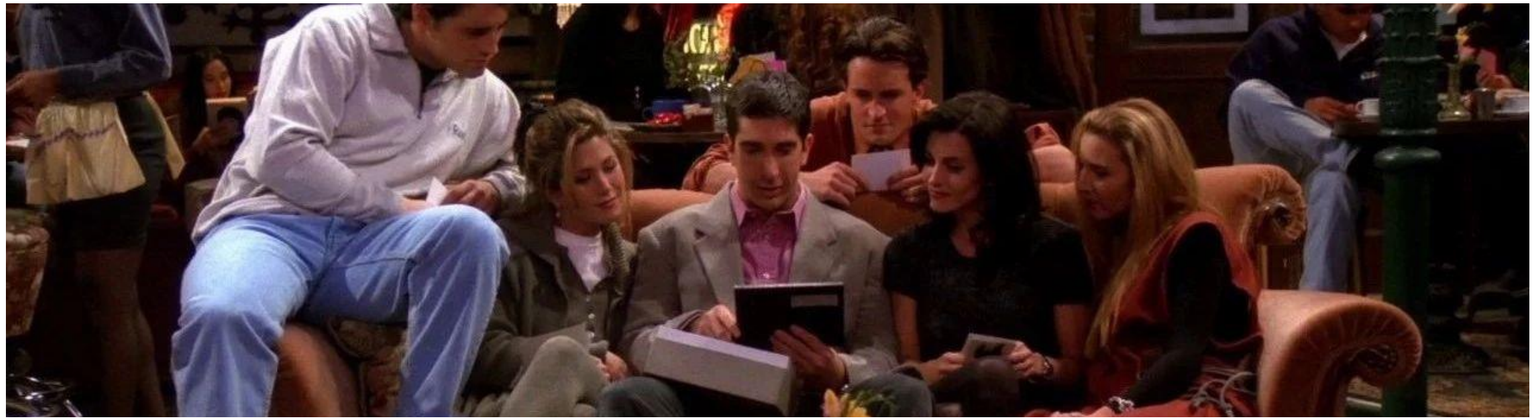
PART 01

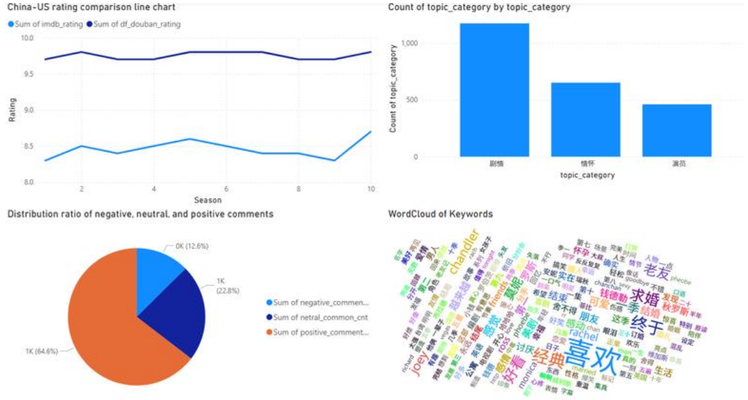
Purpose

AIM OF PROJECT

Purpose

Aim of project





Purpose

- This project aims to quantitatively analyze the acceptance of the classic American drama friends among the Chinese audience.
- The project adopts the whole process technical route of "data collection, natural language processing, multi-dimensional visualization"

Work Flow

Summary of work

Step4:
Data Virtualization

Step3:
Make Model

Step1:
Data Collect

Step2:
Data Extract

PART 02

System configurations

THE DESIGN OF SYSTEM AND BUDGET

System Configuration

Cloud Platform:

Google Colab

Programming Language:

Python

Database:

MongoDB Atlas

Data Processing Framework:

Apache Spark

Machine Learning Model:

Logistic Regression and Naive Bayes

Visualization Tool:

Microsoft Power BI

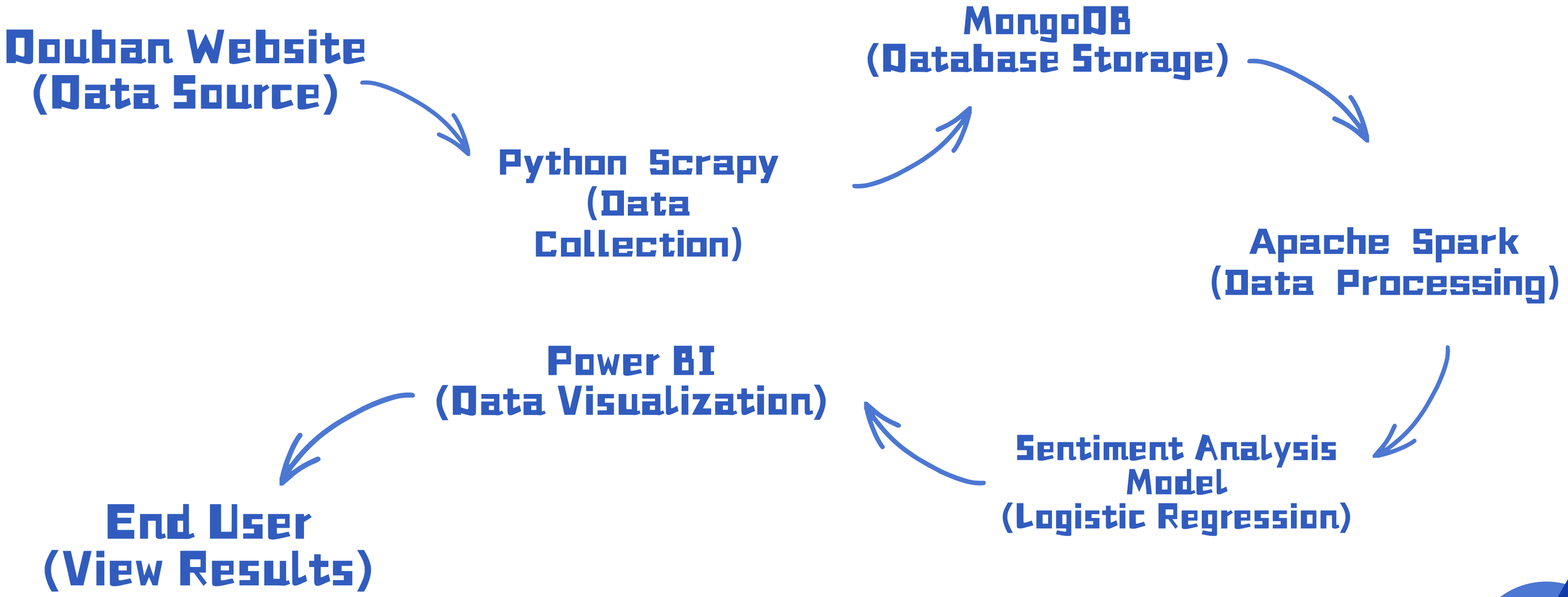
Storage:

Azure Blob Storage

Estimated Data Size:

Approximately 2,000–2,500 user comments

System Architecture of Proposed Big Data System



Budget Plan – Vendor A (Microsoft Azure)

Component	Service	Monthly Cost (HKD)
Virtual Machine	Azure B2s VM (2 vCPU, 4GB RAM)	800
Database	MongoDB Atlas	600
Data Processing	Azure Spark	800
Storage	Azure Blob Storage	200
Visualization	Power BI Pro	100

Total Monthly Cost:
HKD 2,700

Total Yearly Cost:
HKD 32,400

The total cost is well below the HKD 200,000 budget limit and is sufficient to process approximately 2,285 user comments.

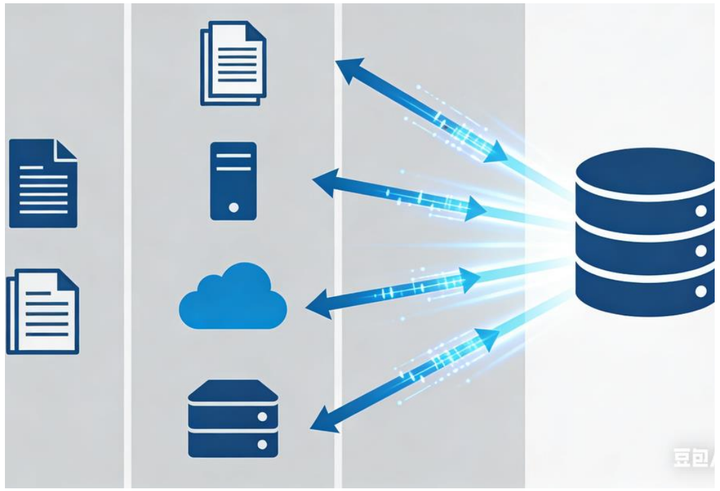
PART 03

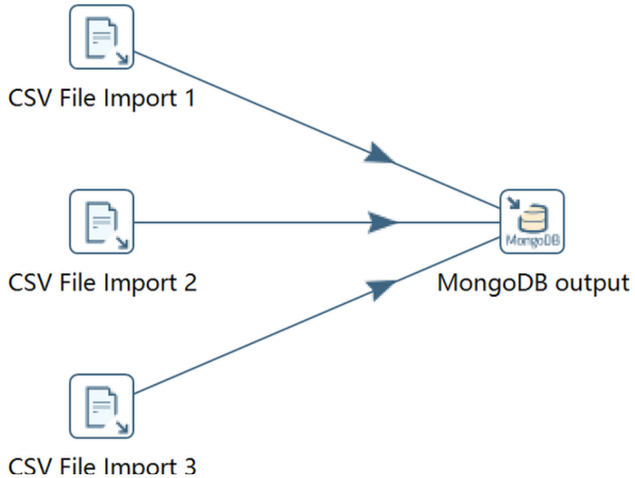
Data Extract

COLLECT DATA-> DATA CLEANING (ODS->DWS)

Collection->ODS->DWS

Data Flow





Collection

We use Python Scrapy to scrapy 400 short comments from douban ,then output as CSV file.

ODS

The full name of ODS is “Operational Data Store”. We import CSV files into the MongoDB-based ODS database, preserving the original data structure, to serve as the foundation for building the DWS layer in the future.

DWS

We build the DWS layer through topic aggregation and data cleaning. Perform data cleaning before summarizing, and then divide the subject fields based on the ODS detail layer data to form the DWS layer subject summary table.

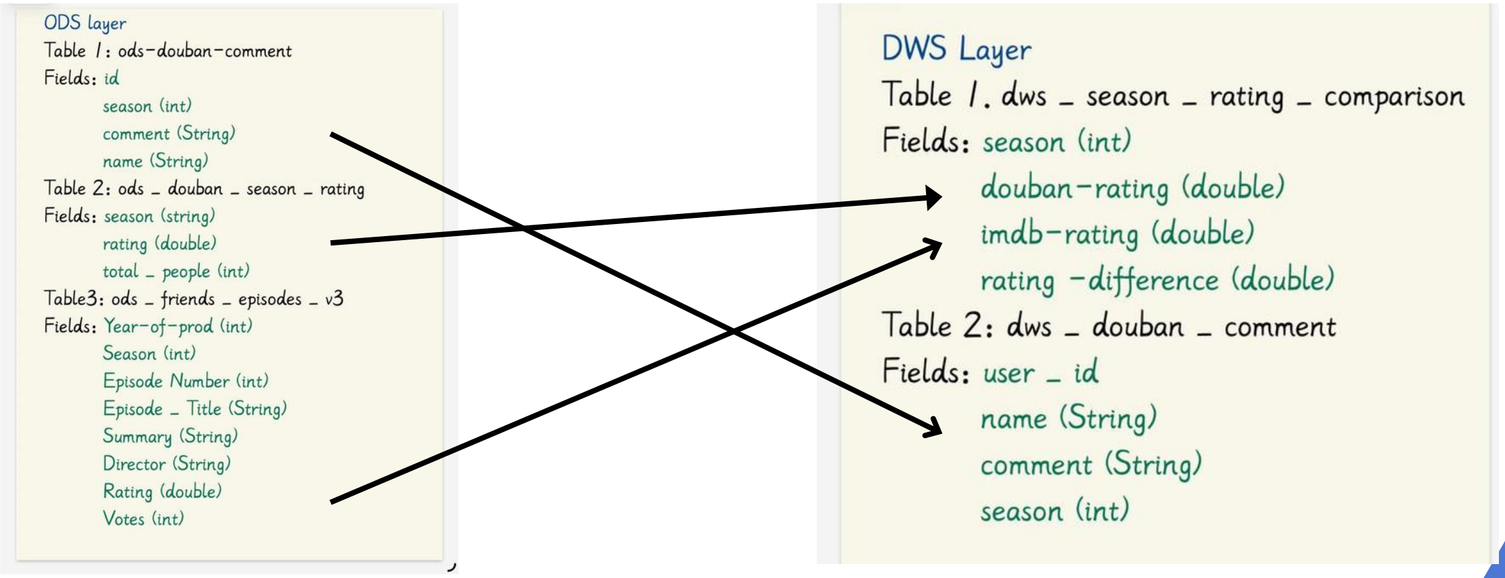
More Detail

Data Cleaning

In this project, during the ETL process from the ODS to the DWS layer, we first perform data deduplication, followed by sorting the records in ascending order based on the TV series season.

Data Subject Division

Based on the field topic of ODS layer, we divide and summarize the data into two core topic tables: rating comparison table and comment table. The first table aggregates rating indicators of each season to quantify audience feedback, and the second table aggregates audience comments of each season. After data cleaning and unifying the indicator caliber, we finally form a standardized and efficient indicator query service in DWS layer, providing solid data support for modeling and analyzing the acceptance of friends in China in ads layer.



PART 04

Make Model

DWS → ADS

Project Overview: Core Functionalities and Value Proposition

Our system acts as the core of data intelligence analysis, transforming unstructured text into valuable, quantifiable business insights through AI processing.

Core Functionalities

Sentiment Analysis

Determines review sentiment (positive/negative/neutral).

Topic Categorization

Categorizes reviews into predefined topics (e.g., cast, plot).

Core Value Proposition

Data Value Enhancement

Enhances data value by structuring unstructured text.

Decision Support

Supports data-driven decisions for product and marketing.

System Architecture: The Seven-Step Analysis Pipeline

An efficient data processing engine based on rule filtering, AI model screening, and LLM arbitration.

Data Preparation

- Rule Filtering: Removes low-quality reviews
- Model Filtering: Identifies & removes spam
- Preprocessing: Cleans text for NLP

Core Analysis

- Sentiment Analysis: Uses multiple models & LLM
- Topic Categorization: Classifies into topics (cast, plot)

Result & Output

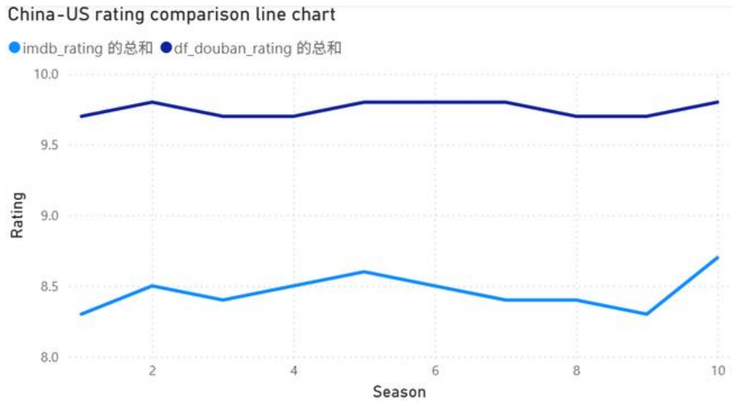
- Result Merging: Unifies sentiment & topics
- ADS Output: Generates data products

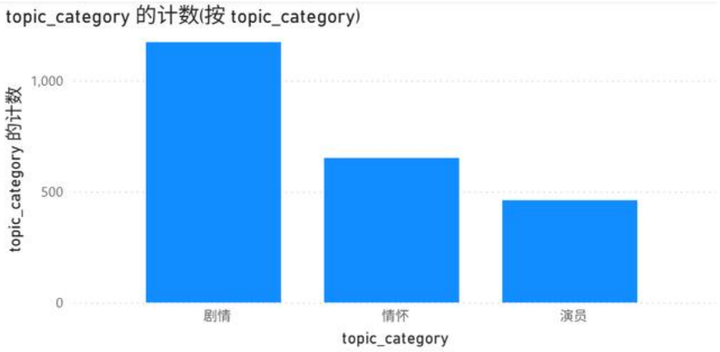
PART 05

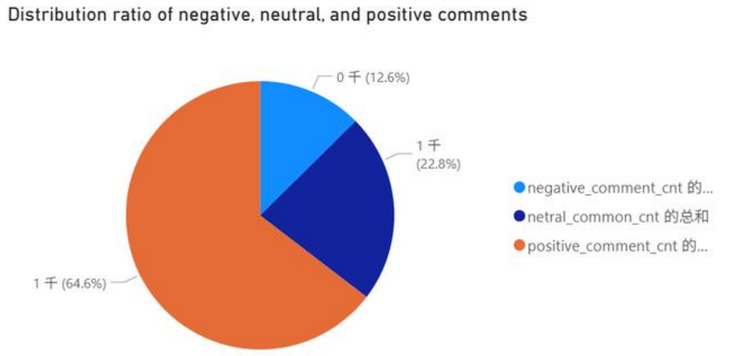
Data Virtualization

EXPLAIN THE DATA

Visualization Dashboard







WordCloud of Keywords
喜欢
经典
好看
求婚
终于
老友
钱德勒
罗斯
莫妮卡
chandler
rachel
joey
monica
ross
phoebe
friends
感动
感觉
感人
越来越
结束
结婚
秋罗斯
朋友
可爱
讨厌
生活
希望
舍不得
眼泪
惊喜
订婚
婚礼
混乱
陪伴
特别
原谅
发现
剧集
半年
像话
婚姻
二十
十年
美国
亮点
设定
维加斯
恭喜
舍得
真的
一刻
第五
标记
重温
集真
表情
心疼
爆笑
性格
married
东西
正果
欢乐
恋爱
日子
真
mon
chan
chanchan
pheebe
goodbye
口语
sexy
不错
第八
轻松
瑞秋
人生
情节
人物
一点
时间
打算
完美
场景
第七
怀孕
问字
字幕
失落
季一
反反复复
一口气
安妮
实在
搞笑
菲比
第十
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伤感
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几集
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2026

THANK YOU

Hu Jihan
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Tang Chao